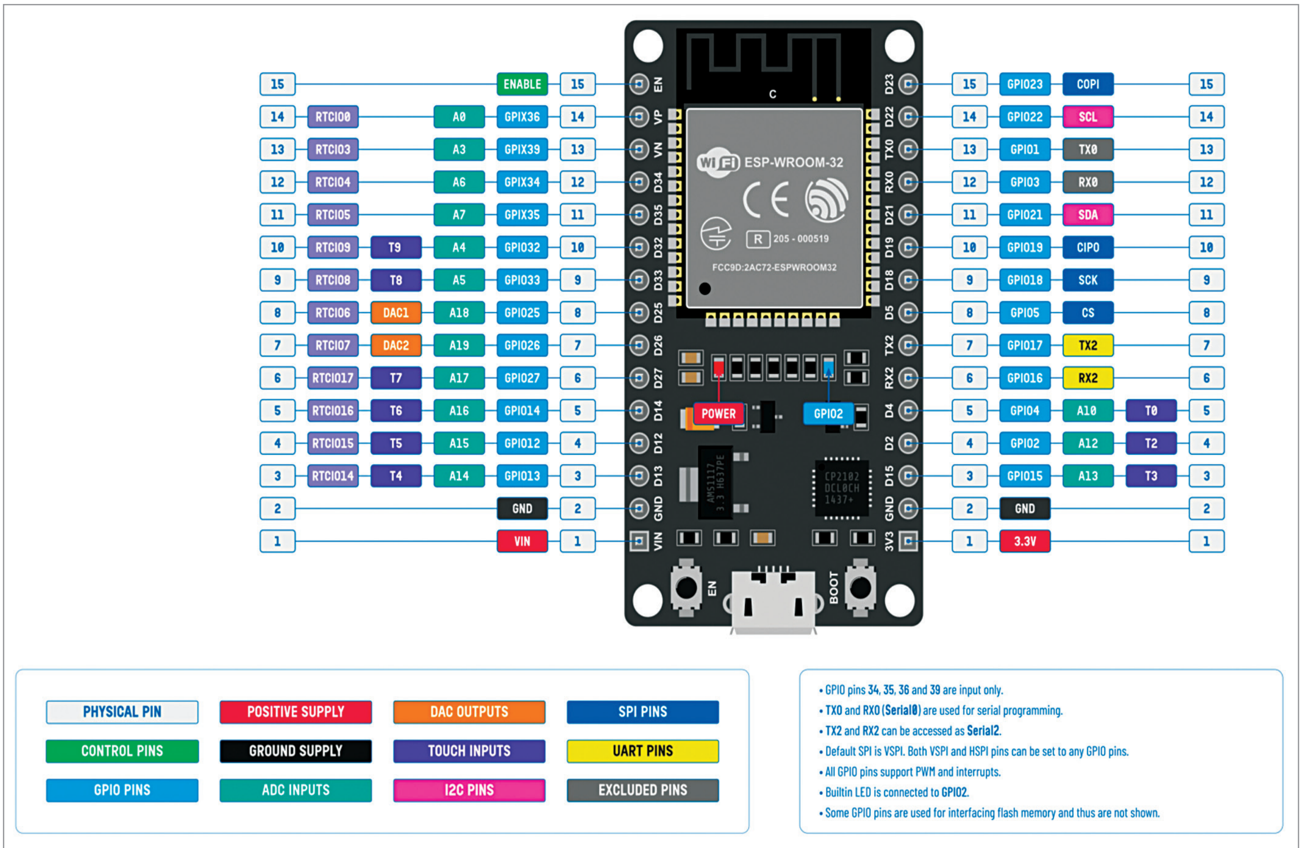


*Fig. 4: The pin diagram of the ESP32 development kit V1 microcontroller*

pH < 7 is acidic, pH > 7 is basic, and pH = 7 is neutral. There are two methods for measuring pH: colorimetric methods utilising indicator solutions or papers, and more accurate electrochemical methods using electrodes and a millivoltmeter (pH meter). We are employing the latter method with the aid of a microcontroller circuit for enhanced accuracy in readings.

Plants have varying pH requirements; however, a pH range of 5.5 to 6.5 is generally considered optimal for hydroponic gardening. Most plants thrive in slightly acidic conditions. The commonly used chemicals to adjust pH are phosphoric acid (to lower pH) and potassium hydroxide (to raise pH). While both chemicals are relatively safe, they can cause burns and should never come in contact with the eyes.

**TABLE 1**
**VARIOUS POWER MODES OF OPERATION AND POWER CONSUMPTION OF ESP32 CHIP**

| Power Mode | Description | | | Power Consumption |
|---|---|---|---|---|
| Active (RF working) | Wi-Fi Tx packet | | | — |
| | Wi-Fi/BT Tx packet | | | |
| | Wi-Fi/BT Rx and listening | | | |
| Modern sleep | The CPU is powered up | 240MHz | Dual-core chip(s) | 30mA~68mA |
| | | | Single-core chip(s) | NA |
| | | 160MHz | Dual-core chip(s) | 27mA~44mA |
| | | | Single-core chip(s) | 27mA~34mA |
| | | Normal speed: 80MHz | Dual-core chip(s) | 20mA~31mA |
| | | | Single-core chip(s) | 20mA~25mA |
| Light sleep | — | | | 0.8mA |
| Deep sleep | The ULP coprocessor is powered up | | | 150µA |
| | ULP sensor-monitored pattern | | | 100µA @ 1% duty |
| | RTC timer + RTC memory | | | 10µA |
| Hibernation | RTC timer only | | | 5µA |
| Power off | CHIP_PU is set to low level, the chip is powered down | | | 1µA |